Detection of Antarctic Surface Meltwater Using Sentinel-2 Remote Sensing Images via U-Net With Attention Blocks: A Case Study Over the Amery Ice Shelf

Lihang Niu, Xueyuan Tang, Shuhu Yang, Yun Zhang, Lei Zheng, and Lijuan Wang

***Abstract*— Surface meltwater critically impacts the Antarctic mass balance and global sea level rise. Quantifying the extent of surface meltwater in Antarctica on a large scale is a challenging task. Traditional methods, such as thresholding, have many limitations. We used a deep learning method, the U-Net with attention blocks, to automatically extract surface meltwater from Sentinel-2 images. We inserted attention mechanism blocks into U-Net to assign different weights to all pixels and channels to utilize the high resolution and multiple channels of Sentinel-2 images. The model was used to map surface water bodies in Sentinel-2 images, and the average accuracy reached 0.9969 on the test dataset. In East Antarctica, the Amery ice shelf (AIS) exhibits the largest surface meltwater area. Studying surface meltwater dynamics on the AIS is useful for understanding the East Antarctic mass balance and demonstrating the model performance. We analyzed the classification results for surface water bodies on the AIS from January 2017 to 2022. Spatially, 96% of surface water bodies are concentrated inland of the AIS from 70° S to 73° S and account for 93% of the region 20 km from the coastline of the AIS. Temporally, the water body area varies considerably in different years, with a maximum in 2017 (932.54 km^2) and a minimum in 2021 (58.34 km^2). The spatial distribution of surface water bodies on the AIS is controlled by the firn air content (FAC), katabatic winds, bare rocks, and blue ice. The interannual variability is associated with complex climate factors, including temperature, surface net solar radiation, snowfall, and snowmelt, among which temperature and snowfall show a strong correlation.**

***Index Terms*— Amery ice shelf (AIS), attention mechanism, deep learning, surface meltwater.**

Manuscript received 16 January 2023; revised 5 April 2023; accepted 8 May 2023. Date of publication 11 May 2023; date of current version 26 May 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 42276257, Grant 41941006, and Grant 41876230. *(Corresponding author: Xueyuan Tang.)*

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Digital Object Identifier 10.1109/TGRS.2023.3275076

I. Introduction

WIDESPREAD distribution of surface meltwater was observed in Antarctica [1]. Meltwater formed at high elevations flows toward ice shelves at lower elevations and accumulates in depressions, forming rivers and lakes [2]. This has important implications for the glacier dynamics and mass balance of the Antarctic ice sheet [3], [4]. First, surface meltwater enters the ice sheet base through fractures, enhancing basal sliding, and accelerating ice flow [5]. This phenomenon has been confirmed on the Antarctic Peninsula [6]. Second, meltwater forms surface runoff on the ice shelf, directly leading to mass loss of the Antarctic ice sheet [5]. Finally, surface meltwater can trigger ice shelf collapse [7]. In the days before the Larsen B ice shelf collapse, a large number of supraglacial lakes were rapidly drained via hydrofracturing, which likely triggered the ice shelf collapse [8], [9]. Collapse leads to lower back stresses of the ice shelf and thus faster ice flow, which increases Antarctic mass loss [9], [10], [11]. The Sixth Assessment Report from Intergovernmental Panel on Climate Change (IPCC) noted that the contribution of Antarctic ice mass loss to the global sea level rise until 2100 under different shared socioeconomic pathway (SSP) scenarios could range from 0.03 to 0.27 m (SSP1-2.6), 0.03 to 0.29 m (SSP2-4.5) or 0.03 to 0.34 m (SSP5-8.5) [12]. Within these contexts, it is essential to obtain the spatial and temporal distributions of Antarctic surface meltwater to better understand the Antarctic contribution to sea level rise.

Remote sensing is a vigorous tool for investigating the distribution of surface water bodies on ice shelves, even across the Antarctic ice sheet. With massive amounts of remote sensing images, researchers have aimed to determine an automated method for surface meltwater analysis. The normalized difference water index (NDWI) entails the use of thresholds to detect water bodies in multispectral satellite images [13], [14]. The NDWI was used to map the distribution of water bodies on East Antarctic ice shelves [4]. Then, cloud, rock, and shadow masks were added [15]. However, the NDWI suffers drawbacks and requires manual correction of errors in the classification of non-water features as water.

Deep learning algorithms have been utilized to detect surface meltwater in the Antarctic. Deep learning involves

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the use of computational models with multiple processing layers to recognize or predict new data by extracting effective representations from large amounts of data through varying degrees of abstraction [16]. Deep learning has been widely applied in Antarctic's research. To solve the challenge of extracting the coastline of Antarctica from Sentinel-1 satellite images, Heidler et al. [17] proposed HED-UNet. Dong et al. [18] proposed the EisNet deep neural network to extract isochronous layers from ice radar data. The application of deep learning in extracting Antarctic surface meltwater is still in its early stages. Dirscherl et al. [19] employed an improved U-Net to distinguish the distributions of Antarctic supraglacial lakes. However, there remain pressing issues that must be addressed in using deep learning to identify Antarctic surface water bodies, such as the construction of highly credible deep learning datasets and the selection of a suitable deep learning model to match the characteristics of remote sensing images.

We used a U-Net model with attention blocks to achieve automated identification of Antarctic surface water bodies from Sentinel-2 multispectral remote sensing images. The tasks addressed in Antarctic research using deep learning mainly fall within the scope of semantic segmentation, including the extraction of surface meltwater in Antarctica. In Earth observation, the U-Net and its variants, which consist of an encoder-decoder architecture, are commonly used for semantic segmentation [20]. Compared to other models, they have better performance [17], [19]. The structure of the U-Net gives it a strong ability to perceive contextual information, which enables it to perform well in semantic segmentation tasks [19]. In addition, compared to other deep learning models, the U-Net has fewer parameters and a faster training speed [21]. Furthermore, because the U-Net can learn more accurate feature representations from small sample data, it performs well in semantic segmentation tasks with small sample data [22]. These characteristics make the U-Net the preferred choice for the task of extracting Antarctic surface meltwater. However, some surface meltwater is located in complex terrain areas, such as mountains and cloud shadows, where the original U-Net struggles to identify them. Moreover, unlike ordinary images, remote sensing images have multiple channels and high resolution. To address these issues, attention blocks were added to the U-Net. The attention mechanism can help the model extract more useful feature information and further improve the performance of semantic segmentation [23].

Supervised learning based on training data is a major training method in the field of deep learning, and more high-quality labeled data are needed as the network depth increases [16], [24]. Antarctic surface water bodies are widely and unevenly distributed. It is a time-consuming and labor-intensive task to manually create labels through visual identification of meltwater in multispectral images. NDWI [15] was used to construct label data, greatly reducing the complexity of dataset creation. In this article, a workflow was proposed to automatically identify Antarctic surface meltwater in Sentinel-2 images, which provides a basis for studying meltwater dynamics on the surface of the AIS. The surface water body distribution and evolution patterns on the AIS from 2017 to 2022 were investigated based on the established model.

II. Data and Methods

A. Study Field

As shown in Fig. 1, the AIS area is approximately 6.26×10^4 km^2, and it is the largest ice shelf in East Antarctica [25]. This ice shelf drains approximately 16% of the mass of the East Antarctic ice sheet, located between 73° S and 69° S beside the Prince Charles Mountains and Princess Elizabeth Land [26]. There is the largest volume of supraglacial lakes of the AIS in East Antarctica [27]. Much time is needed to map the distribution of supraglacial lakes on the AIS because the spatial density is much higher than that in most areas of East Antarctica [4]. Therefore, investigating this region could verify the feasibility and validity of the model proposed in this article. In recent years, the quantification of surface meltwater area on the AIS has been studied as shown in Fig. 2. With the exception of Tuckett et al. [28], most studies focus on the period between 2014 and 2020. During this time, although the exact values differ, they show a similar trend of change. An early study predicted a large-scale collapse of the AIS in 2025 or later [29]. However, in September 2019, the AIS collapsed, which was the largest collapse since 1960 [30]. Knowledge of the spatial and temporal distributions of surface water bodies could help to better understand the effect of surface water bodies on ice shelf collapse. To evaluate the variation in surface meltwater on the AIS in different years, the AIS coastline, as determined by the Scientific Committee on Antarctic Research (SCAR) Antarctic Digital Database [31], was used to delineate the study field to ensure consistency.

B. Data Preparation

Sentinel-2 images are available free of charge from the Copernicus Open Access Hub (available at https://scihub.copernicus.eu/). Sentinel-2 image bands, including three visible bands and a near-infrared band with 10 m resolution, an infrared band and two short-wave infrared bands with 20 m spatial resolution, and other bands with 60 m resolution. All bands were resampled to a 10-m spatial resolution using nearest-neighbor interpolation [34].

Antarctic surface meltwater is widely distributed and dispersed, with many lakes or thin streams of a small size. The size of a given Sentinel-2 image is 10980×10980 pixels, and manual extraction of Sentinel-2 images is a tedious task. The labeled dataset for surface water bodies mainly involves two steps: 1) creating labeled data using NDWI and 2) manually revising the NDWI results. We used the NDWI [15] for coarse extraction of Antarctic surface water bodies from Sentinel-2 images, and approximately 1–2 min were needed to process one Sentinel-2 image. The extraction produced a single-band image with the same resolution as that of the original Sentinel-2 image, and the values of each pixel point include 1 and 0, with 1 indicating a water body and 0 indicating a non-water body.

Although cloud, rock, and shadow were masked out from the NDWI, other interfering features still remained, as shown

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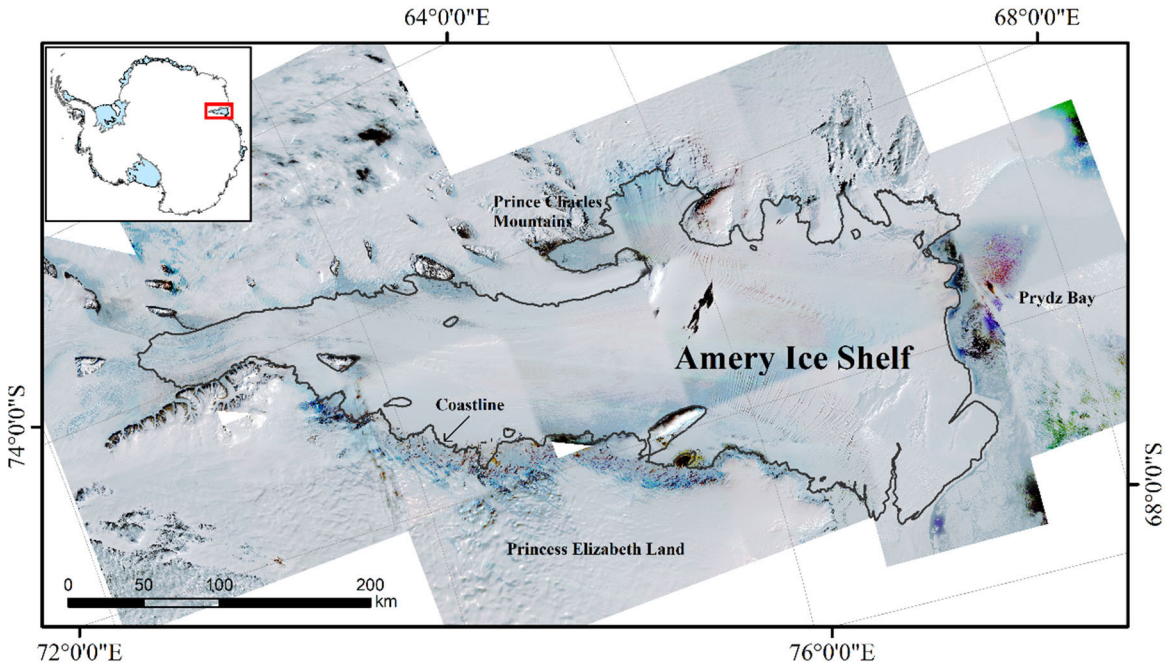


Fig. 1. Study field is the AIS. The illustration shows the location of the study field in the Antarctic. The background image is a true color image assembled from multiple Sentinel-2 images recorded in January 2022. The black lines indicate the coastline of the AIS, as retrieved from the Scientific Committee on Antarctic Research (SCAR) Antarctic Digital Database [22].

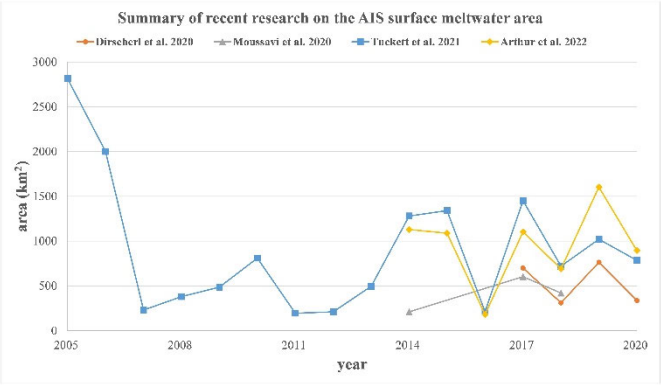


Fig. 2. Summary of recent research on the AIS surface meltwater area. Dirscherl et al. 2020 is [32]. Moussavi et al. 2020 is [33], where data for the years 2015 and 2016 are missing. Tuckett et al. 2021 is [28], Arthur et al. 2022 is [27].

in Fig. 3. First, the dark blue clouds in Fig. 3(a) were incorrectly classified as water bodies [see Fig. 3(b)], and the manual correction result is shown in Fig. 3(c). Next, the lower parts of the uneven surface area [see Fig. 3(d)] were classified as water bodies in Fig. 3(e), and the manually corrected result is shown in Fig. 3(f). Third, thin streams and small lakes near the base rocks in Fig. 3(g) were incorrectly classified as non-water bodies by the shadow mask, and the red circle in Fig. 3(i) marks the correction.

Sentinel-2 images are high-resolution images with 13 bands of data totaling up to hundreds of megabytes in size, which may cause problems such as memory overflow if directly used to train deep learning models. In addition, coarsely labeled images of the original size are inconvenient for manual correction. Due to the limitations of memory size and the convenience and accuracy of manually revising label data, a 128 × 128 grid window was chosen to cut both the coarsely labeled and Sentinel-2 images, which is smaller than previous studies [32], [35]. The window was first moved along the vertical direction and then along the horizontal direction. The segmented Sentinel-2 image with the npy format was then stored as a 128 × 128 × 13 matrix in the file system.

TABLE I
SENTINEL-2 DATA USED FOR CONSTRUCTING THE DATASET

ID	File name	Date	Relative Orbit
1	S2B_MSIL1C_20200119T131859_N0208_R095_T19DEA	Jan 2020	95
2	S2B_MSIL1C_20210131T041729_N0209_R061_T41CPV	Oct 2021	61
3	S2B_MSIL1C_20210131T041729_N0209_R061_T42CVD	Oct 2021	61
4	S2B_MSIL1C_20210131T041729_N0209_R061_T42DVF	Oct 2021	61

A total of 969 samples were generated using the above method from the Sentinel-2 data listed in Table I, of which 80% was used for training and 20% for testing. The training dataset was augmented with vertical, horizontal, and diagonal flips, totaling 2908 samples. The test dataset comprised 242 samples in all.

C. Model Training

A U-Net model with an attention mechanism was proposed to identify Antarctic surface water bodies. The U-Net

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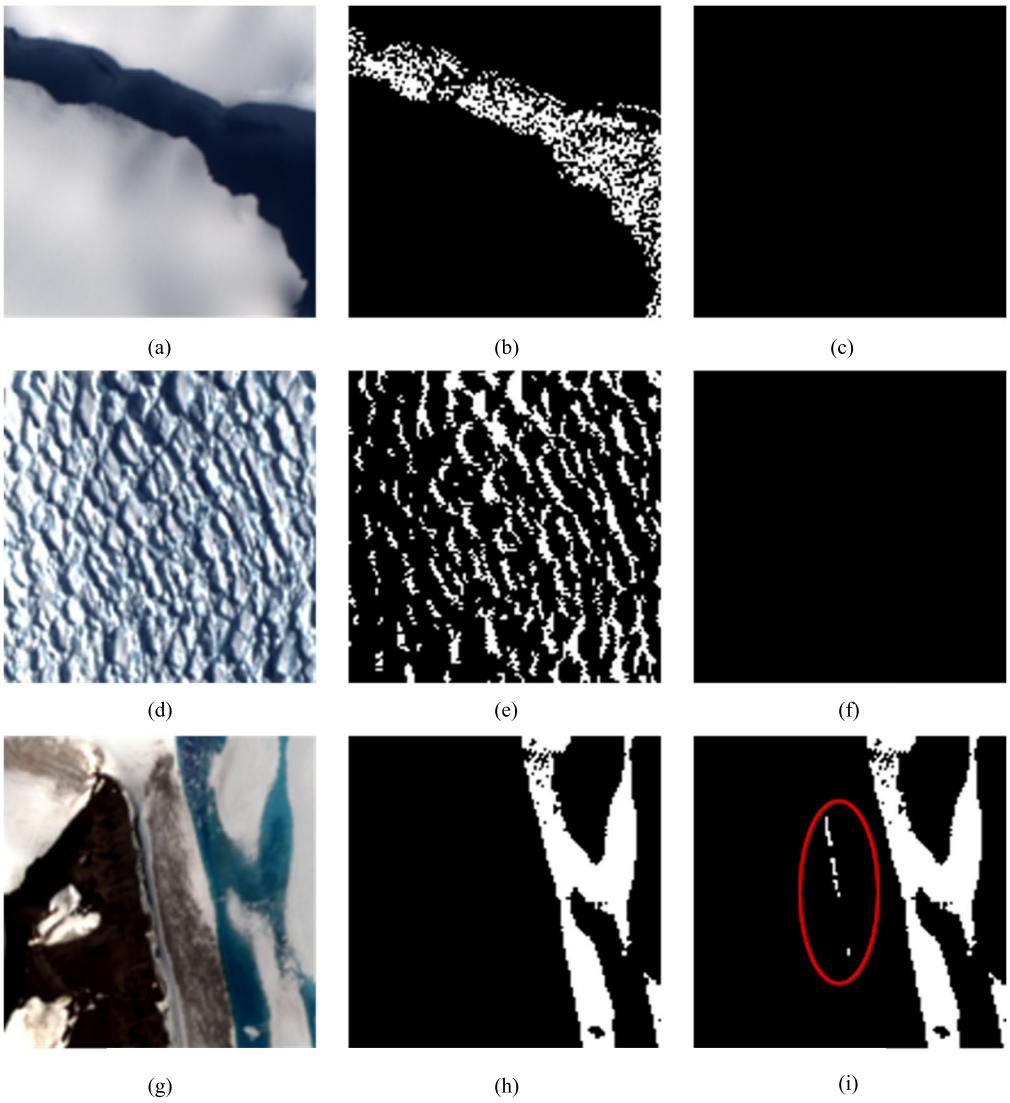


Fig. 3. Comparison between the NDWI classification results and manual correction results. Panels (a), (d), and (g) show Sentinel-2 true color images. The corresponding NDWI classification results are shown in (b), (e), and (h). The manual correction results are shown in (c), (f), and (i), respectively. The white pixels indicate values of 1, which are water bodies. The black pixels indicate values of 0, which are non-water bodies.

model is a fully convolutional neural network that can obtain satisfactory semantic segmentation results on small training datasets [21], [36]. U-Net is composed of two parts, the encoder and the decoder. The image height and width are reduced, and the number of channels gradually increases in the encoder. In the decoder, the image height and width are restored to the original values, and the number of channels is reduced. In addition, the jump connections in U-Net can effectively solve the problem of gradient disappearance [21]. U-Net and U-Net-like structures have been applied to Antarctic research. Zhao et al. [37] used a modified U-Net model to automatically extract Antarctic surface fractures. Barlow et al. [38] used U-Net to extract the boundaries of the fast ice flow zone in complex regions.

The attention mechanism is another important component of the neural network proposed in this study. It was first proposed in 2014 for natural language processing and has since been adopted in a wide variety of fields, especially in computer vision, with excellent performance [39], [40], [41]. The process of observing data in the human brain inspired the attention mechanism [23]. The attention mechanism assigns different weights to different parts of the considered data, suppressing irrelevant content and highlighting features useful for specific tasks [42]. The attentional mechanism can be divided into spatial attention and channel attention aspects. The convolution block attention module (CBAM) contains both spatial attention and channel attention blocks and can be integrated into any convolutional neural network [43]. The efficient channel attention network (ECA-Net) solves the paradox between performance and complexity in the attention block, using a few parameters to significantly improve performance [44].

The structure of an attention block is shown in Fig. 4(b). In order to ensure size alignment between inputs k and g, an upsampling layer followed by a 3×3 convolution and ReLU activation is employed to match the size of k with that of g.

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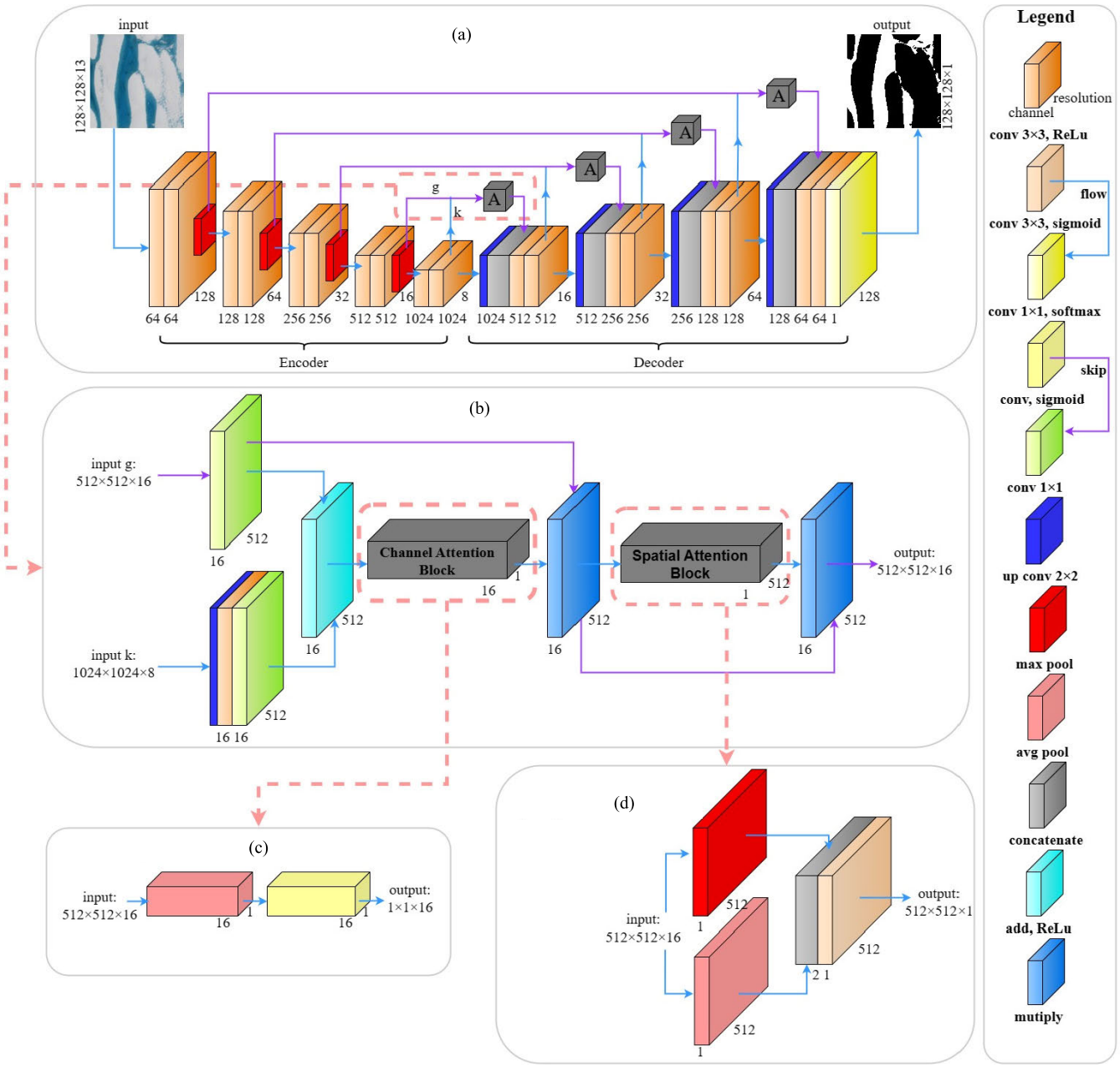


Fig. 4. Structure of U-Net with attention blocks, where (a) is the main part of the neural network, (b) describes the structure of the attention block, (c) is the channel attention block, and (d) is the spatial attention block, and the relationship among the four parts is represented by dashed connecting lines.

The two parts of the input are summed after 1×1 convolutions and then pass through the ReLU activation function, after which they enter the channel attention block. The output is multiplied by the input g and enters the spatial attention block. The output of the spatial attention block is multiplied by the output of the previous layer to obtain the result of the attention block.

Among them, the structure of the channel attention block is shown in Fig. 4(c), which is derived from ECA-Net and comprises a mean pooling layer, a 1-D convolution process, and a sigmoid activation function. The 1-D convolution process of ECA-Net entails the use of an adaptive convolution kernel size, which can be calculated with the following formula:

$$k = \left| \frac{\log_2 C}{\gamma} + \frac{b}{\gamma} \right|_{\text{odd}}. \tag{1}$$

The output of the channel attention block is the weight of each channel. As shown in Fig. 4(d), the spatial attention block comprises a maximum pooling layer, a mean pooling layer, and a 3×3 convolution process using the sigmoid function as the activation function, which is structured from the CBAM spatial attention block. The output of the spatial attention block is the weight of each pixel.

The model was developed based on the PyTorch framework [45]. The training parameters include 200 epochs and 40 batch size, with a learning rate of 0.0001.

D. Assessment Methods

The model results were evaluated via the formulas below. The true positive (TP) value indicates the number of correctly classified water pixels, the false-positive (FP) value indicates

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the number of classified water pixels that are actually non-water pixels, the false-negative (FN) value is the number of classified non-water pixels that are actually water pixels, and the true negative (TN) value is the number of correctly classified non-water pixels. With these indicators, the following can be obtained:

$$\mathrm{PA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{2}$$

PA is the pixel accuracy, determined as the ratio of the number of correctly classified pixels to the total number of pixels

$$P_{\mathrm{water}} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{3}$$

$$P_{\mathrm{water-free}} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FN}} \tag{4}$$

P_{water} is the ratio of correctly classified water pixels to the total number of pixels, and $P_{\mathrm{water-free}}$ is the ratio of correctly classified non-water pixels to the total number of pixels

$$\mathrm{IoU}_{\mathrm{water}} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{5}$$

$$\mathrm{IoU}_{\mathrm{water-free}} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FN} + \mathrm{FP}}. \tag{6}$$

IoU denotes the ratio of the intersection and concatenation of the model classification and true results for a given category

$$\mathrm{MIoU} = \frac{\mathrm{IoU}_{\mathrm{water}} + \mathrm{IoU}_{\mathrm{water-free}}}{2}. \tag{7}$$

MIoU is the average of all-category IoU value

$$K = \frac{\mathrm{PA} - \mathrm{EA}}{1 - 0}. \tag{8}$$

K denotes the Kappa coefficient [46], which is an evaluation criterion to compare the accuracy between the model and the random classifier. Kappa coefficient (K) is usually more representative than the accuracy when evaluating a model. A Kappa coefficient value close to 1 indicates a close approximation to the true result. EA denotes the expected accuracy, which can be calculated with the following formula, and TS indicates the total number of pixels:

$$\mathrm{EA} = \frac{(\mathrm{TN} + \mathrm{FP}) \times (\mathrm{TN} + \mathrm{FN}) + (\mathrm{FN} + \mathrm{TP}) \times (\mathrm{FP} + \mathrm{TP})}{\mathrm{TS} \times \mathrm{TS}}. \tag{9}$$

III. Results

A. Classification Results

Regarding the problems associated with the NDWI classification results shown in Fig. 3, which still have interference features, the neural network model could obtain better results of the defects, as shown in Fig. 5. In Fig. 5(a) and (b), clouds and gullies were corrected by the neural network as non-water bodies [see Fig. 5(c) and (f), respectively], which were previously classified as water bodies in the NDWI results [see Fig. 5(b) and (e), respectively]. In Fig. 4(h), the NDWI could not identify narrow supraglacial rivers due to rock and cloud shadows [see Fig. 5(g)]. The neural network could more accurately distinguish these features. The areas improved by the neural network are marked with red circles in Fig. 5(i).

Fig. 6 shows the neural network classification results of surface water bodies. The water bodies in Fig. 6(a) are located in regions with flat topography and fewer disturbance features. The classification results in Fig. 6(b) show that the water bodies are small, with the largest reaching 2 km and the smallest less than 1 km. Fig. 6(c) shows the Central Terminal Lake (CTL), which is the largest seasonal lake on the AIS [47]. The classification result in Fig. 6(d) shows that the model could identify the CTL, which exhibits an elongated shape with a length of approximately 40 km and a width of approximately 3 km. With a dense distribution in Fig. 6(c), the CTL and surrounding small water bodies could be identified. Fig. 6(e) shows the area with a rough surface, and the distribution of water bodies is scattered. As shown in Fig. 6(f), all the water bodies in the area could be successfully identified. The model could identify the water bodies in Fig. 6(g) situated near the mountains. The result depicted in Fig. 6(h) reveals that the model could accurately distinguish rocks and water bodies.

TABLE II
Comparison of the Assessment Results of U-Net With Attention Blocks With Other Models on the Test Dataset

Methods	FCN	SegNet	U-Net	U-Net with attention blocks
PA	0.9102	0.9815	0.9912	**0.9969**
P_{water}	0.7131	0.8926	0.9013	**0.9172**
$P_{water-free}$	0.9354	0.9821	0.9939	**0.9968**
IoU_{water}	0.6243	0.8530	0.8859	**0.9082**
$\mathrm{IoU}_{water-free}$	0.8731	0.9723	0.9868	**0.9954**
MIoU	0.7487	0.9127	0.9364	**0.9518**
K	0.6626	0.8705	0.8960	**0.9018**

B. Assessment Results

We compared the performance of U-Net with attention blocks against other models on the test dataset, including fully convolutional network (FCN) [48], SegNet [49], and the original U-Net [21], all of which are well-known models in semantic segmentation. The results are summarized in Table II. The neural network developed in this article achieved improved performance across all evaluation metrics. In terms of the various evaluation metrics, FCN showed a significant gap compared to other models. Compared to SegNet and the original U-Net, U-Net with attention blocks improved the overall accuracy by 1.6% and 0.6%, respectively. The accuracy of water body identification was improved by 2.8% and 1.8%, and the IoU was improved by 6.5% and 2.5%, respectively. The accuracy of non-water body identification was improved by 1.5% and 0.3%, and the corresponding IoU was improved by 2.4% and 0.9%, respectively. The MIoU of U-Net with

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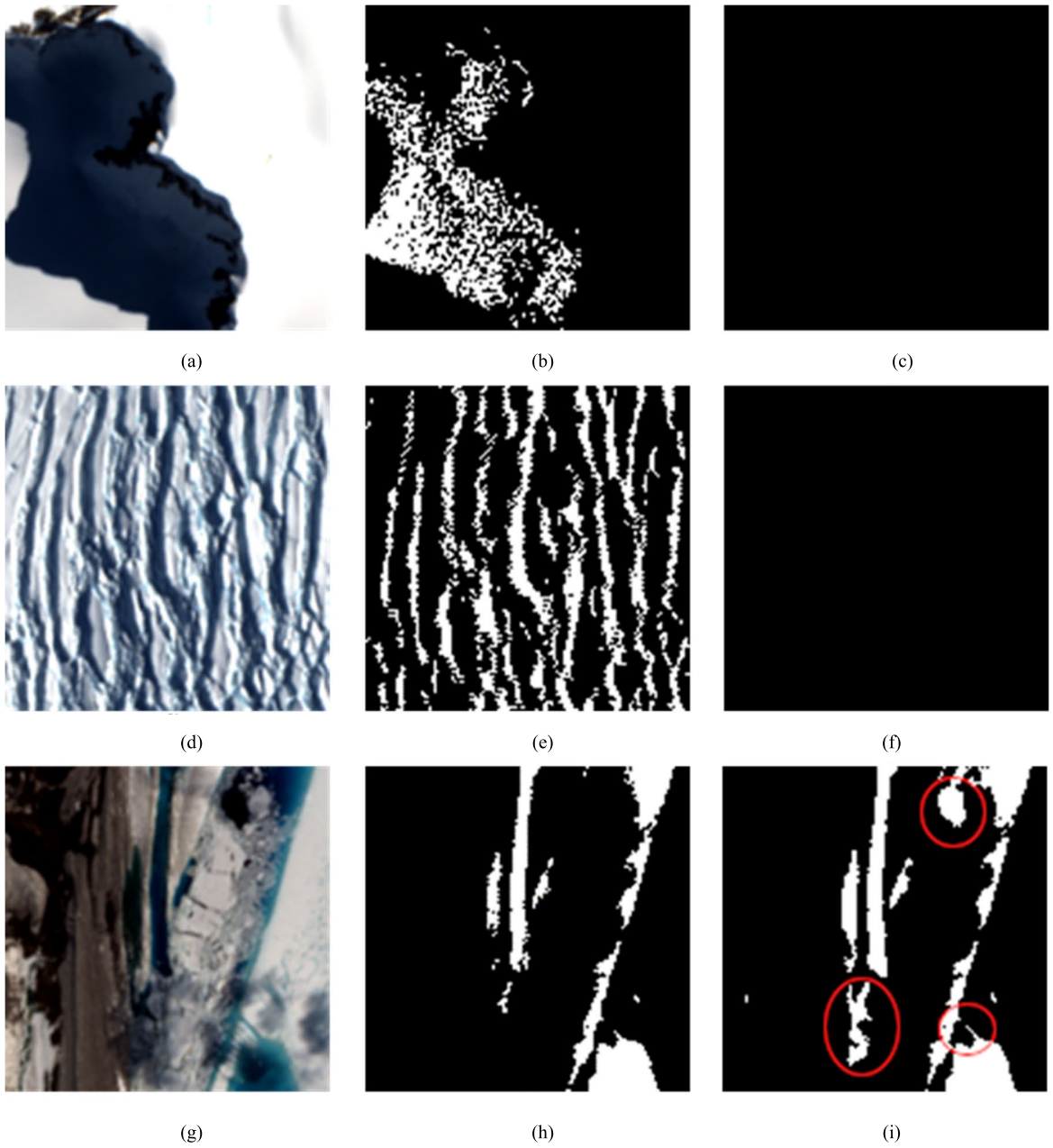


Fig. 5. Comparison of results between the NDWI and model classifications. Panels (a), (d), and (g) show Sentinel-2 true color images. Panels (b), (e), and (h) show the classification results using the NDWI. Panels (c), (f), and (i) show the classification results using the model. In the classification results, the black pixels indicate non-water bodies, and the white pixels indicate water bodies.

attention blocks reached 0.9518, an improvement of 4.3% and 1.6% compared to SegNet and the original U-Net, respectively. The Kappa coefficient reached 0.9018, an improvement of 3.6% and 0.6%, respectively.

C. Spatial Distribution

Considering the good performance of U-Net with attention blocks, we investigated the dynamics in AIS surface water bodies for January 2017–2022 using the model. By definition, the water body and non-water body values are 1 and 0, respectively. The pixel values were summed to produce the recurrence frequency of surface water bodies, as shown in Fig. 7. The maximum frequency value is 6. The water bodies were mainly distributed from 70° S to 73° S. Except for some small water bodies, the supraglacial lakes mostly exhibited an elongated shape, especially near 72° S, where a large lake with a length of approximately 40 km and a width of approximately 3 km occurred. The surface water bodies were concentrated near the Clemence Massif and Fisher Massif. In the region from 73° S to the front edge of the ice shelf, the surface water bodies were mainly concentrated in two areas, one located near the Prince Charles Mountains and the other located near the coastline of the AIS and Princes Elizabeth Land. Water bodies in these areas are relatively small and exhibit a high frequency.

D. Temporal Evolution

Because the actual area corresponding to one pixel is 100 m^2, the surface water body areas were determined in

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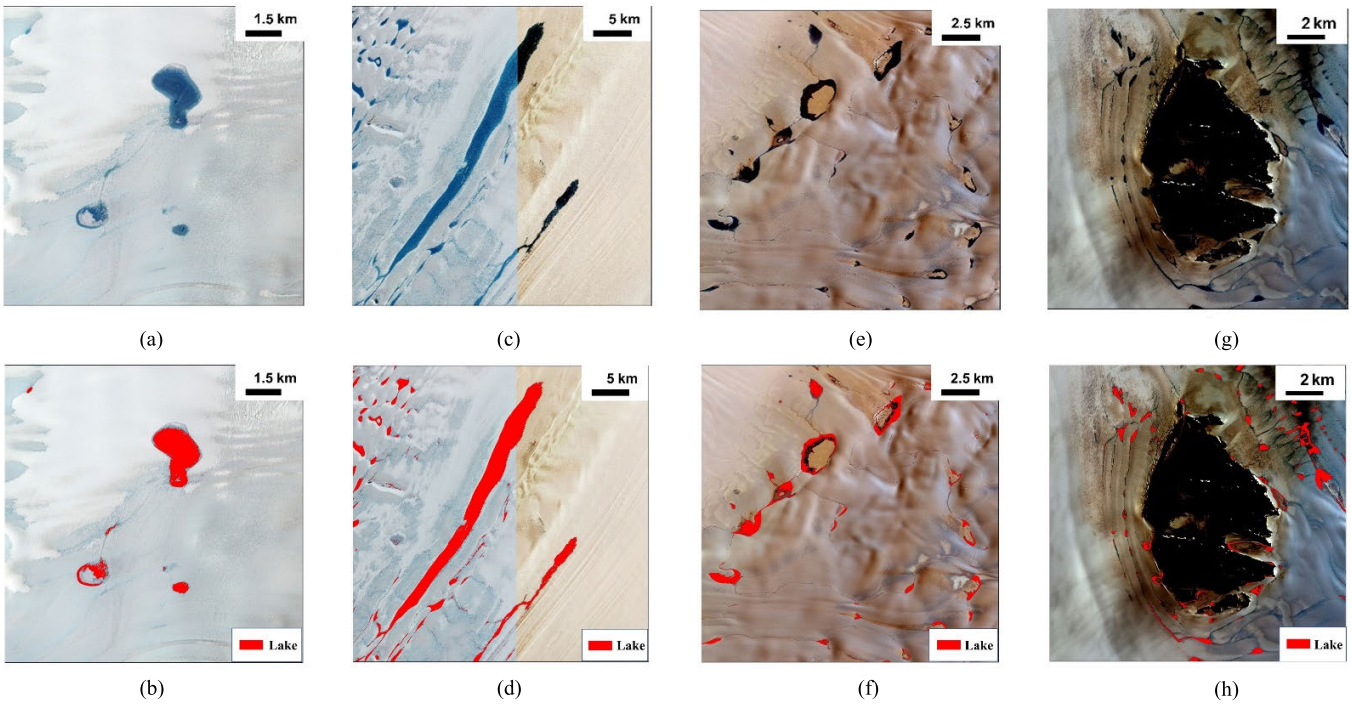


Fig. 6. Neural network model for Sentinel-2 images (a), (c), (e), and (g) to automatically extract ice lake results (b), (d), (f), and (h).

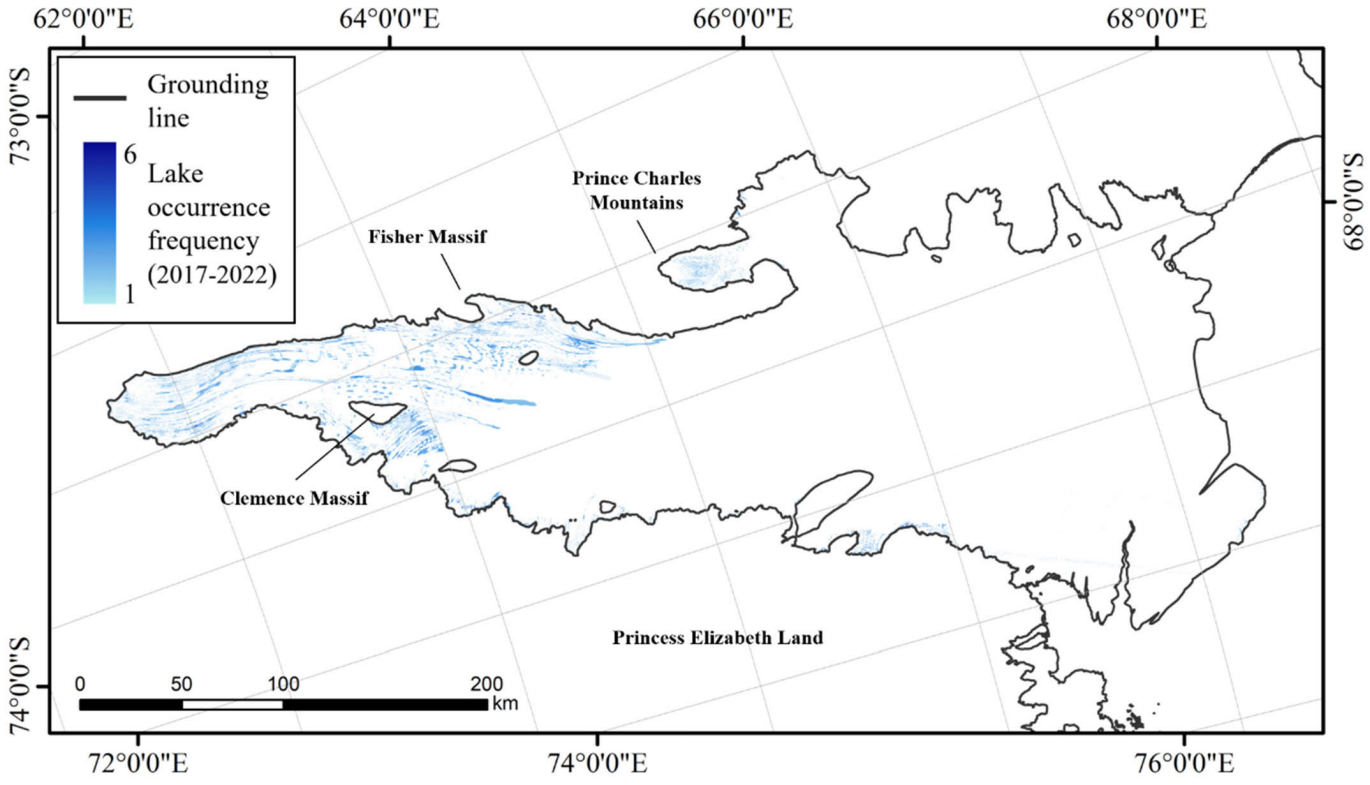


Fig. 7. Recurrence frequency of AIS surface water body, 2017–2022.

different years, as shown in Fig. 8. Due to the Sentinel-2 revisit cycle and cloud cover occlusion, it is hard to obtain all data on a particular day. The Sentinel-2 images used in this study captured the period from January 1 to 31. From January 2017–2022, the peak area of the surface water bodies on the AIS occurred in 2017 (932.54 km^2), and the area in 2021 was the smallest (only 58.35 km^2). Comparing the two periods of 2017–2019 and 2020–2022, the surface water body

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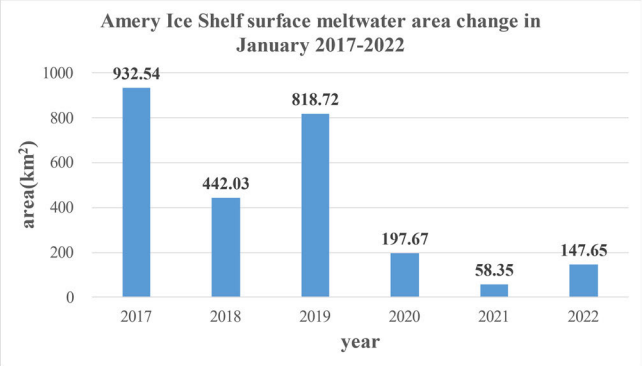


Fig. 8. Change in the area of surface water bodies on the AIS from January 2017 to 2022.

area during the former period was generally larger than that during the latter period. During these periods, the maximum and minimum area values differed by 53.6% (2017 and 2018) and 70.5% (2020 and 2021), respectively.

IV. Discussion

A. Classification Results

In this article, the promising results of the U-Net model with attention blocks for the classification of surface water bodies in the Antarctic were described. Under different underlying surfaces, the model could effectively suppress disturbance features and accurately classify water bodies. The model classification results were also accurate for the large supraglacial lakes mentioned in the literature, such as the CTL. Sentinel-2 images of the AIS in successive years were also classified, and the results demonstrated that the model could effectively monitor the dynamic changes in surface water bodies.

The NDWI involves manual processing to correct for false-positive results such as those depicted in Fig. 4(b) and (e). The results of the model proposed in this study automatically resolved these errors without manual processing [see Fig. 4(c) and (f), respectively]. Based on this finding, some surface water bodies that could not be accurately identified by the NDWI were further classified [see Fig. 4(i)]. Comparing the results of the surface water for the AIS in 2017, a surface water body area of approximately 893.3 km^2 [15] could be derived based on the NDWI, while our result was 932.54 km^2, an increase of 4% over the former value, mainly due to the surface water bodies masked by rocks and clouds that could not be identified by the NDWI.

B. Uncertainty Assessment

According to an analysis of the assessment indexes (Table II), the accuracy rate of water body identification is lower than that of non-water body identification. This suggests that some surface water bodies could not be identified. The classification result of Fig. 9(a) is shown in Fig. 9(b). The model failed to completely identify water bodies. It may be due to the complex underlying surface, where water bodies are mixed with refrozen water bodies, resulting in a poor distinction between water bodies and their surroundings, leading

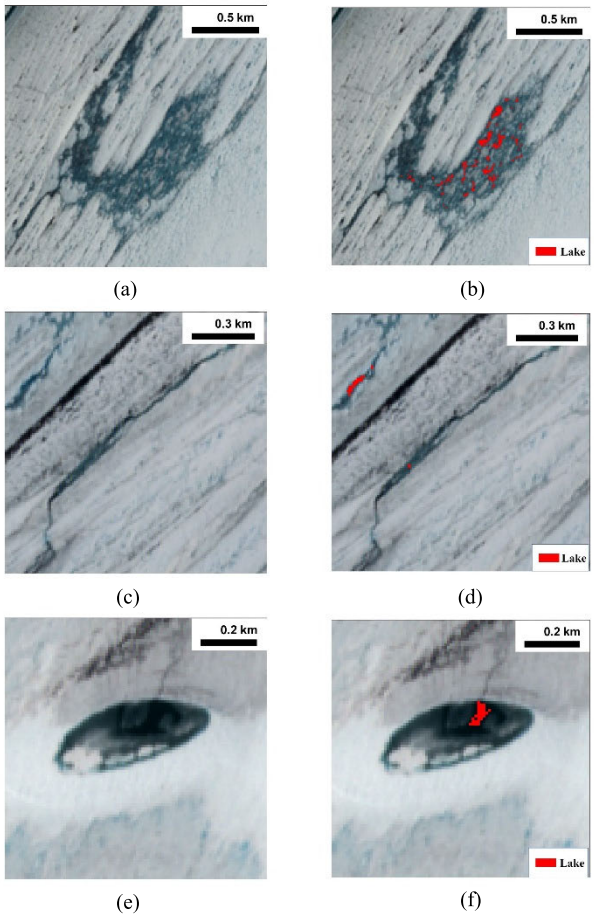


Fig. 9. Cases show the misclassifications in the mapping of surface water bodies from Sentinel-2 images. Panels (a), (c), and (e) show Sentinel-2 images. Panels (b), (d), and (f) show the classification results.

to the model's inability to accurately distinguish water bodies from ice. The recognition result in Fig. 9(c) is not satisfactory, as shown in Fig. 9(d). The reason may be that the narrow water bodies are similar in appearance to surface fractures. In this case, the model incorrectly considered these water bodies as disturbance features. In Fig. 9(e), some regions of the surface water bodies are darker, probably due to the presence of depressions on the surface. Based on the classification result in Fig. 9(f), the model may assume that the darker color of the water bodies is due to shadow effects and therefore may incorrectly identify water bodies in non-depressed areas as interference features. Due to the small size of these problematic surface water bodies, the impact is not notable, accounting for only 0.5% of the total area of the test dataset.

C. Spatial-Temporal Distribution

Here, we present the spatial distribution pattern of surface water bodies on the AIS from January 2017–2022 in Fig. 7. In these six years, the surface water body area within 20 km of the coastline accounted for 93% of the total area on average, and the average percentage in the 70° S–73° S region reached 96%. The surface water bodies are concentrated in the interior of the ice shelf and rarely occur at the front edge near the

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ocean. There is a high accumulation rate at the front edge of the AIS, resulting in a high firn air content (FAC) [50]. A high melting rate and a low accumulation rate occur in the interior, with a low FAC [28]. This may explain the lack of surface water bodies in the front of the ice shelf and the widespread distribution of supra-glacial lakes in the interior of the AIS [51]. When the FAC is high, there is enough space in the snow to store meltwater, and when the FAC is low, snow is rapidly saturated with meltwater, causing meltwater to accumulate on the surface [52]. In addition, the elevation change around the AIS is notable (1000–2500 m), and the slope is steep. The lower elevation of the ice shelf (<1000 m) and gentle terrain make the AIS heavily influenced by katabatic winds [53], [54]. Katabatic winds destroy the inverse temperature near the coastline and raise the humidity [55]. Meanwhile, there are massive blue ice areas (BIAs) due to erosion by katabatic winds [56]. Surface meltwater undergoes refreezing and forms BIAs [57]. A previous study reported that the melting rate of BIAs is twice that of non-BIAs [58]. On the one hand, the albedo of BIAs is much lower than that of snow. On the other hand, BIAs provide impenetrable surfaces for meltwater pooling.

The interannual changes in the AIS surface water body area from January 2017 to 2022 were different (see Fig. 8). The AIS exhibits a high interannual variation in the surface water body area. In order to investigate the climatic drivers affecting the AIS surface meltwater, the relationship between the surface water body areas and the Southern Annular Mode (SAM) index and EAR5 climate reanalysis data was analyzed. SAM index from [59] for the summers from 2016/2017 to 2021/2022 was selected. EAR5 data from [60], which includes various climate variables, such as 2 m temperature (T_{2m}), surface net solar radiation (S_{rad}), snowfall, and snowmelt, were also selected. Among them, the mean values of T_{2m}, S_{rad}, and snowmelt from December to February were used, and the mean value of snowfall in winter, from February to December, was used. These EAR5 data were extracted using a manually drawn mask around the AIS coastline.

The results of Pearson correlation analysis between each climate factor and the AIS surface meltwater areas are shown in Table III. Due to the limited data used for correlation analysis, the p-values may not be representative, and therefore, the focus should be on interpreting the correlation coefficient r. The highest positive correlation was found with T_{2m} ($r = 0.65$, $p = 0.16$), which suggests that as the temperature increases, the formation of the AIS surface meltwater becomes more extensive. This is consistent with the findings of the study on East Antarctica [27], [61] and on the Antarctic Peninsula [62]. SAM index also reflects temperature changes in Antarctica to a certain extent, but it is negatively correlated with the AIS surface meltwater areas. Because, the East Antarctic temperature is usually low when the SAM index is positive, and it is high when the SAM index is negative [63]. There is a positive correlation between snowmelt and the AIS surface meltwater areas, which is quite evident. However, given the lack of evaluation for ERA5 snowmelt products specifically for the Antarctic, this relationship should be interpreted with caution. S_{rad} and the AIS surface meltwater areas have a positive correlation, possibly due to increased incoming shortwave radiation causing more snowmelt and contributing to the formation of surface meltwater [58], [64]. Winter snowfall showed a strong negative correlation with the AIS surface meltwater areas ($r = -0.88$, $p = 0.02$), which may be due to its provision of air storage capacity for summer melting [65]. More snowfall replenishes the firn pore space, increases the FAC, and reduces the accumulation of meltwater on the AIS surface.

TABLE III

RESULTS OF PEARSON CORRELATION ANALYSIS BETWEEN EACH CLIMATE VARIABLE AND THE AIS SURFACE MELTWATER AREAS, WHERE THE PEARSON CORRELATION COEFFICIENT (r) AND THE CORRESPONDING p-VALUE INDICATING STATISTICAL SIGNIFICANCE ARE REPORTED. THE SIGNIFICANCE LEVELS ARE DENOTED AS $*$ FOR $p < 0.05$, $**$ FOR $p < 0.01$, AND $***$ FOR $p < 0.001$

Climatic Variables	r	p
T_{2m} (°C)	0.65	0.16
Snowmelt (m w.e)	0.45	0.37
S_{rad} (W m^{-2})	0.44	0.38
SAM	-0.55	0.27
Snowfall (m w.e)	-0.88*	0.02

D. Future Improvements

A highly reliable dataset plays an important role in monitoring surface water body changes in the Antarctic via deep learning. Due to the lack of high-quality datasets, we used the NDWI to generate a dataset via manual correction. However, this limits the ability of the neural network because the model inherits the defects of the NDWI. A high-quality dataset should cover as many different underlying surfaces as possible, such as water bodies located in mountainous terrain and under cloud shadows. Different shapes and areas of surface water bodies should also be included in the dataset. In addition, to mitigate interfering features as much as possible, rocks, clouds, and BIAs should be labeled. The model could identify a wider variety of image features. In this way, the influence of interference features on surface water body identification could be fundamentally suppressed, and the identification accuracy of surface water bodies could be improved.

The use of multiple sources of data to monitor surface changes is better than the use of a single data source [66]. The use of Sentinel-2 images to study Antarctic surface water bodies is intuitive, and surface water bodies can be accurately identified through visual inspection. However, Sentinel-2 images are severely affected by cloud cover. For part of the time, Sentinel-2 images cannot monitor the Antarctic surface below clouds. Sentinel-1 SAR images do not suffer similar defects, and they can facilitate all-day monitoring of the Antarctic surface. The joint application of multisource data in the identification of Antarctic surface water bodies is a possible choice in the future.

V. CONCLUSION

Here, we used U-Net with attention blocks to automatically identify surface water bodies in Antarctica. Specifically, spatial

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and channel attention blocks were inserted into U-Net by combining the high-resolution and multichannel features of Sentinel-2 images. Corresponding weights were assigned to each pixel and each channel so that the model could better fit the Sentinel-2 images and improve the identification results of the surface water body. We also applied the NDWI to produce a dataset with high confidence and manually corrected for false-positives in the NDWI results.

The trained neural network could effectively suppress interference features and accurately identify surface water bodies in different environments. Overall, the model achieved a pixel accuracy of 0.9969 and a Kappa coefficient of 0.9018. The factors limiting the identification results include: 1) the dataset was produced by the NDWI, and the model, therefore, inherited the defects of the NDWI; and 2) the single data source could not provide more information on the topographic features of the ice shelf to the model.

To validate the reliability of the U-Net with attention blocks, Sentinel-2 images of the AIS surface meltwater from January 2017 to 2022 were identified using the neural network. Spatially, surface water bodies were mainly distributed in the interior and at the coastline of the AIS but rarely occurred at the front edge. Temporally, the AIS exhibited a high variation in the surface water body area. During the study period, the maximum value occurred in 2017 (932.54 km^2) and the minimum value occurred in 2021 (58.35 km^2). The spatial distribution is controlled by the FAC, katabatic winds, bare rocks, and BIAs. The interannual variability of the AIS surface meltwater areas is associated with complex climate factors, including temperature, surface net solar radiation, snowfall, and snowmelt, among which temperature and snowfall show a strong correlation, indicating that the AIS surface meltwater will intensify with increasing temperature and decreasing snowfall.

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